

Association between diabetes and thiamine status - A systematic review and meta-analysis

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ABSTRACT

Background: Thiamine (vitamin B1) is an essential cofactor in glucose metabolism, but it remains unclear whether thiamine status is lower in individuals with diabetes compared to individuals with normal glucose metabolism. **Aims:** We conducted a systematic review and meta-analysis to study whether the circulating concentrations of various thiamine analytes differ between people with and those without diabetes.

Methods: PubMed and the Cochrane Central Register of Controlled Trials were searched according to the study protocol. The standardized mean difference (SMD) and 95 % confidence intervals (CI) of thiamine markers between individuals with and without diabetes were used as effect size (random effects model). Subgroup analysis considered albuminuria as an additional variable.

Results: Out of the 459 articles identified, 24 full-texts were eligible for the study, 20 of which qualified for the data analysis and four were evaluated for coherence. Compared to controls, individuals with diabetes showed lower concentrations of thiamine (pooled estimate SMD [95 % CI]: $-0.97 [-1.89, -0.06]$), thiamine monophosphate ($-1.16 [-1.82, -0.50]$), and total thiamine compounds ($-1.01 [-1.48, -0.54]$). Thiamine diphosphate ($-0.72 [-1.54, 0.11]$) and erythrocyte transketolase activity ($-0.42 [-0.90, 0.05]$) tended to be lower in persons with diabetes than in controls without reaching statistical significance. Subgroup analysis showed that individuals with diabetes and albuminuria had lower thiamine levels than the controls ($-2.68 [-5.34, -0.02]$).

Conclusions: Diabetes is associated with lower levels of various thiamine markers, suggesting that individuals with diabetes may have higher thiamine requirements than those without diabetes, but well-designed studies are required to confirm these findings.

1. Introduction

Recent decades have witnessed a large rise in global diabetes prevalence resulting in profound effects on quality of life, burden on health care, and economic costs [1]. According to the International Diabetes Federation (IDF) Diabetes Atlas, an estimated 537 million people had diabetes in 2021, and this number is projected to reach 783 million by 2045 [2]. Substantial morbidity and increased mortality are attributed to diabetes-related macrovascular complications such as myocardial infarction or stroke and microvascular complications such as blindness, kidney failure, or amputations in people with diabetes [3].

Oxidative stress that diverts upstream metabolites from glycolysis

into pathways of glucose overutilization is an important mechanism involved in the pathogenesis of diabetes complications [4]. Excess accumulation of these metabolites could be due to deficiency of thiamine, an essential cofactor for several enzymes involved in glycolysis and Krebs cycle, suggesting a role of this vitamin in the development of diabetic microvascular complications [5]. The clinical manifestations of thiamine deficiency include neurologic or cardiovascular disease (CVD).

Thiamine is an essential water-soluble vitamin required for mitochondrial energetics, such as the production of adenosine triphosphate (ATP) where it serves as a critical and rate-limiting cofactor to multiple enzymes involved in energy metabolism. Some of these enzymes operate at the entry points and at critical junctures for glucose, fatty acid, and

Abbreviations: CC, (case control); CVD, (cardiovascular disease); PPP, (pentose phosphate pathway); RCT, (randomized clinical trials); SMD, (standardized mean difference); TDP, (thiamine diphosphate); THTR, (thiamine transporters); TMP, (thiamine monophosphate); TTP, (thiamine triphosphate).

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amino acid pathways [5,6]. Once absorbed, free thiamine is phosphorylated into its active form thiamine diphosphate (TDP) (synonym: thiamine pyrophosphate [TPP]) [5,6]. Thiamine has a very short half-life and a limited storage capacity. In addition, it is susceptible to degradation and depletion by a number of products including environmental and pharmaceutical chemicals [6]. Pork, fish, many nuts and seeds, beans, peas, tofu, brown rice, whole wheat, acorn squash, and asparagus are examples of sources rich of thiamine in the diet [6]. The recommended daily allowance (RDA) for thiamine is 1.1 mg and 1.2 mg for adult females and males, respectively. It has been suggested that the current RDA requirement may be insufficient to meet the demands of modern living, since thiamine deficiency is prevalent in multiple patient populations (between 20 % and > 90 %) [6].

Thiamine and its derivative benfotiamine have been shown to counteract diabetic microangiopathy in animal models and some clinical trials [5,7,8]. Thiamine also modulates neuronal and neuromuscular transmission in vertebrates, and deficiency thereof may cause severe nervous system disorders such as beri-beri and Wernicke-Korsakoff syndrome [5]. While the metabolically active TDP constitutes some 80 % of total body thiamine, the analysis of TDP concentrations in whole blood is considered a useful biomarker of thiamine status [9]. Additional phosphates can be added or subtracted from the molecule to form thiamine triphosphate (TTP) or thiamine monophosphate (TMP), respectively. However, the advantages of measuring circulating TTP and TMP in blood are not evident [6,9]. The concentrations of thiamine, TMP, TDP, TTP, and erythrocyte transketolase activity in plasma or blood are widely used as laboratory markers of thiamine status. These markers have been compared between people with diabetes and control individuals, but the results varied considerably. For example, some [10–13] but not all studies [14–18] reported markedly lower concentrations of one or more thiamine markers in individuals with diabetes compared to those without diabetes. We hypothesized that people with diabetes have lower thiamine than those without diabetes. To test this hypothesis, we performed a systematic review and meta-analysis of existing studies.

2. Methods

The study was conducted according to an a priori protocol that was registered at the international prospective register of systematic reviews (PROSPERO, CRD42022306734). The systematic review was conducted according to the PRISMA guidelines.

2.1. Study population, exposure, outcome and inclusion and exclusion criteria

The study population included individuals with diabetes (either type 1 or type 2) and control persons. The exposure is the presence of diabetes and the outcome is thiamine status as shown by at least one of the following markers: whole blood thiamine, erythrocyte thiamine, erythrocytes transketolase activity index (α ETK), erythrocyte transketolase activity, urinary thiamine, or any of the thiamine forms measured in serum, plasma or whole blood [i.e., thiamine (excluding mono- and diphosphate forms), TMP, TDP, total thiamine (the sum of all thiamine forms)].

Inclusion criteria comprised studies published in English or German language and reporting blood thiamine concentrations measured by any assay method. Cross sectional, case control (CC), cohort studies, and baseline data of interventional studies (before starting the intervention) were eligible. Letters to the editor were included only if fulfilling the inclusion criteria and containing the data needed. To be eligible, studies had to include ≥ 50 participants (individuals with diabetes and controls in total). The studies must have shown thiamine concentrations or enzyme activity in participants with diabetes and controls separately. The diagnosis of diabetes could be based on health register data, hospital archive, standard laboratory tests (fasting glucose, HbA1c), use of

diabetes medications, or self-reported diabetes. Studies in individuals with alcohol consumption ≥ 100 g ethanol per week [cutoff for alcohol overconsumption according to Wood et al., 2018 [19]] (if reported) were not qualified. The control group had to be apparently healthy without renal insufficiency. Multiple publications from the same cohort were all included only when showing different thiamine markers.

We excluded studies in women with gestational diabetes (due to higher requirement of thiamine during pregnancy that could cause changes in the biomarkers or induce urinary excretion), studies without or with inappropriate control groups, studies not reporting concentrations of systemic thiamine biomarkers, studies including exclusively people with Wernicke-Korsakoff syndrome, studies including individuals treated with thiamine (any form and dose), case reports, case series, review articles, abstracts of conferences or meetings, and genetic disorders of thiamine deficiency or thiamine responsive inherited disorders (i.e., Rogers syndrome; hereditary autosomal recessive disorder characterized by megaloblastic anemia, diabetes mellitus, and sensorineural deafness). In addition, we excluded studies in participants who underwent bariatric surgery that could be the primary cause of thiamine deficiency, and those with primary gastrointestinal disorders and dementia that could affect thiamine absorption and intake, respectively. When duplicate publications were encountered, we prioritized the publication with the largest sample size and most recent publication date.

Studies were not excluded due to missing co-variables or effect modifiers (i.e., age, sex, duration of diabetes, HbA1c) or due to low quality of the study if otherwise the inclusion criteria are fulfilled.

2.2. Search and screening strategy

The study investigators are specialized in internal medicine, diabetology, and endocrinology (DZ), neurology (KR), biology (AS), and clinical chemistry and epidemiology (RO). The literature search was conducted on 09.02.2022 in PubMed and the Cochrane Central Register of Controlled Trials databases. In addition to hand search, the reference lists of previous systematic reviews and the articles citing the identified articles were screened for potentially relevant studies.

Two authors (DZ and RO) conducted the search using the search strings shown in Supplemental Table 1. The two authors independently performed the screening steps (title and abstract and full text screening), data extraction into a study specific excel sheet, and risk of bias assessment. A third author (KR) was consulted when disagreement during this process occurred. Disagreements were resolved through discussions also with a third author (KR) if needed.

We extracted the following data into the excel sheet; first author, publication year, PubMed identification number, country of participants, study setting (community, primary, secondary, or tertiary care, or national registry), study design, mean (standard deviation, SD) of age and BMI, sex (% F or M), inclusion and exclusion criteria of the original studies, diabetes type, duration of diabetes, HbA1c, plasma glucose, diabetes medications, (% smokers), renal function (mean creatinine, estimated glomerular filtration rate or the presence of albuminuria), diabetic complications whenever reported (i.e., % with chronic kidney disease, retinopathy, polyneuropathy, or CVD), the exact thiamine biomarker(s) measured and the unit of measurement, and the analytical methods used for measurements. The numerical results of interest were the number of the participants with diabetes and the controls and the mean of thiamine marker and SD; mean and standard error (SE); or mean and 95 % confidence intervals (95%CI) in each of these groups.

For publications that principally obtained the data but were published more than 10 years ago (before January 2012), we approximated the mean and SD from available sources such as figures. For studies published after January 2012 but did not publish the data in the required form, we contacted the authors at least twice to obtain the data.

We assessed the risk of bias for case-control and cross-sectional studies using a personalized version of the Newcastle-Ottawa scale. In

addition, the risk of bias of individual studies was assessed at the study level (internal validity of the study) using an individualized tool that evaluates several points in study design such as selection of participants, confounding factors, and missing information [20].

2.3. Statistical analysis

The meta-analysis was run for each thiamine marker separately. We undertook a meta-analysis when the same thiamine marker was reported by at least 3 independent studies. The meta-analysis was undertaken at the study level. Thus, for studies presenting mean (SD) values among subgroups of participants (i.e., with and without albuminuria) that shared the same control group, we assumed that the correlation coefficient between the subgroups of a study equal to 1.0.

The meta-analysis aggregates the results from studies contrasting thiamine markers in individuals with diabetes versus the controls in the same study. The point estimates of the synthesized standardized mean difference (SMD) and the confidence intervals are reported. We used the SMD in each study as the numerical result to be meta-analyzed. SMD is the difference in thiamine concentrations or transketolase activity levels between individuals with diabetes and those without diabetes relative to the variability observed in that study (the common SD). The SMD can be used regardless of the actual scales used to make the measurements. The SMDs were estimated by Cohen's *d* and Hedges' *g*, a bias corrected estimator. The random-effects model was used since the analyses are based on a random sample of the studies identified and because the SMDs vary between the studies. We used Forest plots to present the results of the individual studies and the pooled estimates.

Two independent studies by Al-Daghri et al., assessed two distinct groups from the same population-wide survey including participants with type 1 diabetes [11] and those with hyperglycemia and the metabolic syndrome [21]. However, we rated the SDs reported by Al-Daghri et al., 2013 [11] for all thiamine markers as being implausibly low causing falsely high SMD values. The corresponding author did not respond to our repeated inquiries. Therefore, to minimize this bias and avoid exclusion of this study [11], we imputed the SDs from their second publication, where the same design and assay methods were used [21]. Accordingly, we assumed an SD of 2.0 for thiamine, 1.0 for TMP, 15.0 for TDP, and 15.5 ng/ml for total thiamine. We also assumed equal SDs for subjects with diabetes and controls. The original uncorrected results are presented in the online resources for transparency.

We studied heterogeneity using the I-squared (I^2) statistics that quantify the proportion of the observed variance that reflects real difference in the effect size rather than sampling error. Publication bias (e.g. due to unpublished studies or studies published in languages other than English) was studied by visual inspection of asymmetry of the Funnel plots of the individual SMDs on the x-axis versus the standard error on the y-axis. If publication bias was suspected, a trim-and-fill method was used to correct for the bias by assuming a hypothetical SMD for the missing studies to be replaced.

A leave-one-out meta-analysis was run to investigate whether removing any specific study from the analysis would have a major effect on the pooled estimate which may invalidate the conclusions. Subgroup analysis included analysis according to renal dysfunction. We originally planned to run subgroup analysis according to co-morbidities [i.e. retinopathy, CVD, neuropathy, albuminuria]. However, the number of studies providing thiamine markers in subgroups was not sufficient to run a meta-analysis.

We calculated the prediction intervals (PIs) to study the variability of the effect between studies. The PI of the SMD is calculated from the point estimate of SMD, its standard error (SE) and the estimated tau-square (τ^2). If we can assume that the effects are normally distributed (in the relevant units), we can expect that the true effect size and 95 % of all comparable populations will fall within this interval [22].

The quantitative data analysis was performed for all available studies independent of the quality of the study. The data analyses were

conducted using Comprehensive-Meta-Analysis Software program (Biostat Inc., Version 3). *P* values <0.05 were considered to be statistically significant.

3. Results

3.1. Search results

The initial search identified 459 potentially relevant articles. After the first screening wave of abstracts and titles, 430 articles were excluded due to exclusion criteria and 29 articles qualified for the full text screening. After excluding 5 articles, the remaining 24 full-texts were included in the systematic review. The excluded articles and the reasons for their exclusion are shown in Supplemental Table 2. Twenty out of the 24 articles qualified for the quantitative data analysis and additional 4 were qualitatively evaluated for coherence with the results of the quantitative analysis (Fig. 1, Study Flow Diagram). Table 1 includes a brief description of the studies identified and those included in the statistical analysis. The majority of the studies specified thiamine markers as thiamine, TMP, TDP, transketolase activity or coefficient or total thiamine, while seven studies specified measuring serum or plasma thiamine [10–15,21]. Supplemental Table 3 summarizes thiamine markers and analytical methods as reported in the original studies.

We had access to the dataset of Nix et al., and thus we were able to provide mean (SD) of the different study groups. Nine authors were contacted for additional data, but only 2 [14,23] provided the data; 1 declined to provide the data, and 6 authors did not reply after 2 additional reminders.

3.2. Association between diabetes and thiamine markers

3.2.1. Presence of diabetes and thiamine concentrations

We identified seven independent studies that assessed thiamine concentrations in plasma [12–14], serum [11,15,21], or whole blood [10]. The SMD with 95%CI for the 7 studies and the pooled estimate are shown in Table 2. Four studies reported lower thiamine concentrations in participants with diabetes compared to those without diabetes [10–13], while two studies showed no difference between the two groups [14,21] and one study reported higher serum concentrations of thiamine in the diabetes group compared to the controls [15]. The pooled analysis showed lower thiamine concentrations in individuals with diabetes than in the control groups [pooled SMD (95%CI) = -0.97 ($-1.89, -0.06$); $n = 7$ studies, $I^2 = 97.62$, $P < 0.001$; SE = 0.95, $\tau^2 = 1.47$]. The prediction interval ranged from -4.31 to 2.37 , indicating that the true effect in 95 % of all comparable populations will fall within this interval. The Funnel plot of the standard error (SE) on the y-axis by SMD for thiamine concentrations suggested no publication bias (*p* value for Egger's regression intercept = 0.096). (Supplemental Fig. 1A). Removing the studies one by one did not exert a major effect on the SMD (Supplemental Fig. 1B).

We conducted a subgroup analysis according to the presence and the absence of albuminuria in individuals with diabetes compared to the controls, respectively. Seven studies reported thiamine concentrations in subgroups of people with albuminuria [10,12–15], while the same studies reported the concentrations in subgroups of individuals without albuminuria. The subgroup analysis showed diabetes with albuminuria to be associated with lower thiamine concentrations compared to the controls [SMD (95%CI) = -2.68 ($-5.34, -0.02$)]. Individuals with diabetes without albuminuria tended to have lower thiamine concentrations than the controls without reaching statistical significance [SMD (95%CI) = -0.99 ($-2.28, 0.30$)] (Fig. 2).

3.2.2. Presence of diabetes and concentrations of thiamine monophosphate

Five independent studies reported concentrations of TMP in plasma, serum or whole blood of individuals with and those without diabetes [10,11,13,15,21] (Fig. 3). All studies reported lower mean TMP

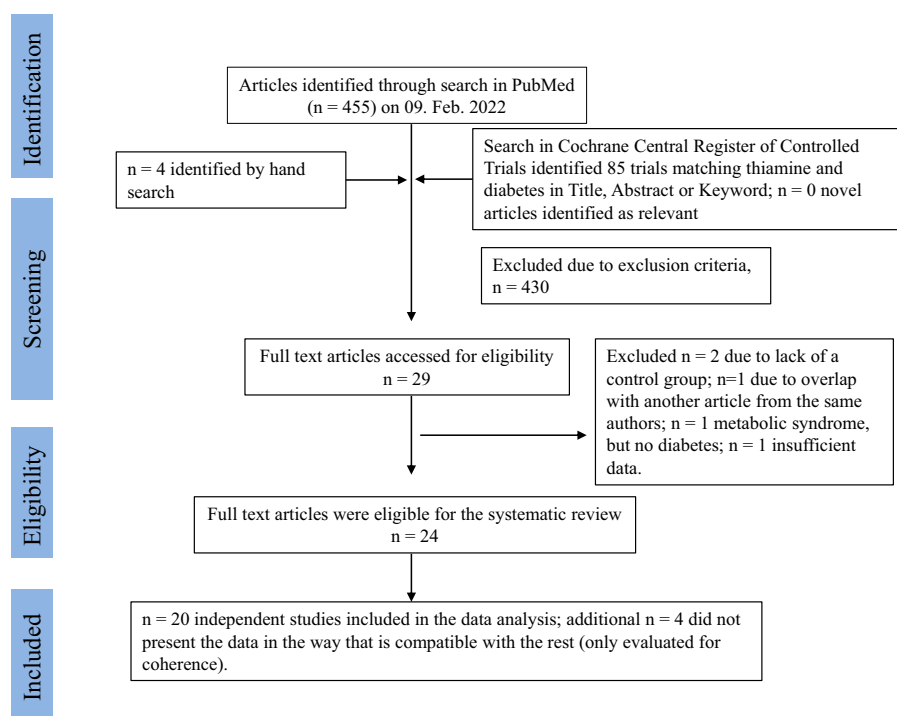


Fig. 1. Study Flow Diagram.

concentrations in individuals with diabetes compared to controls. The pooled analysis showed lower TMP concentrations in individuals with diabetes in comparison to controls [SMD (95%CI) = -1.16 (-1.82 , -0.50), $n = 5$, $I^2 = 96.62$, $P < 0.001$; $\text{Tau}^2 = 0.54$]. The predictive interval ranged from -3.73 to 1.41 .

The studies assessing TMP showed no publication bias ($p = 0.102$ and Funnel Plot of SE by SMD). (Supplemental Fig. 2A).

Removing one study from the meta-analysis each time did not markedly change the overall SMD (Supplemental Fig. 2B).

3.2.3. Presence of diabetes and concentrations of thiamine diphosphate

Nine studies reported concentrations of TDP in individuals with diabetes and control individuals [10,11,14–18,21,23]. TDP was measured in serum, whole blood or red blood cells. The presence of diabetes was not associated with lower concentrations of TDP compared to the controls [SMD (95%CI) = -0.72 (-1.54 , 0.11), $n = 9$ studies, $I^2 = 98.04$, $p < 0.001$; $\text{Tau}^2 = 1.54$] (Table 3). The predictive interval ranged from -3.82 to 2.38 . The studies showed no publication bias (p value for Egger's regression intercept = 0.387). The pooled SMD according to Duval and Tweedie' trim and fill method was -0.69 (-0.80 , -0.59) (Supplemental Fig. 3A).

Removing one study from the meta-analysis each time did not markedly affect the overall SMD values (Supplemental Fig. 3B).

3.2.4. Presence of diabetes and concentrations of total thiamine

We identified four studies reporting concentrations of total thiamine in serum [11,21] or whole blood [10,24]. All studies were consistent in finding lower concentrations of total thiamine in individuals with diabetes compared to the controls [SMD (95%CI) = -1.01 (-1.48 , -0.54); $n = 4$ studies; $I^2 = 87.63$, $p < 0.001$; $\text{Tau}^2 = 0.20$] (Table 3). No publication bias was detected (p value for Egger's regression intercept = 0.387) (Supplemental Fig. 4A). The prediction interval ranged from -4.92 to 2.90 .

When the studies were removed one by one, the overall SMD showed consistently lower total thiamine in subjects with diabetes (Supplemental Fig. 4B).

3.2.5. Presence of diabetes and transketolase activity and transketolase activity coefficient

We identified five studies reporting levels of transketolase activity [14,25–28] and six reporting transketolase activity coefficient in red blood cells [26,28–32]. The transketolase activity is a direct measure of the residual enzyme activity, while the transketolase activity coefficient estimates the TPP effect. Thus, lower thiamine status will be associated with lowering of the transketolase activity and an increase in the transketolase activity coefficient. The pooled SMD estimate for the transketolase activity coefficient did not differ between individuals with diabetes and controls (Fig. 4 A).

While two studies reported lower transketolase activity in the diabetes group, three found no differences between the diabetes and control groups (Fig. 4 B). The pooled SMD showed a non-significant trend toward lower levels in individuals with diabetes than in controls [-0.42 (-0.90 , 0.05)]. The corresponding Funnel Plots (Supplemental Fig. 5A and B) did not suggest publication bias for transketolase activity or transketolase activity coefficient.

3.2.6. Qualitative evaluation of studies or substudies not included in the statistical analysis

We identified four novel studies that could not be included in the quantitative data analysis due to a lack of appropriate data [33–36]. In addition, in 5 studies we were unable to meta-analyze all thiamine biomarkers, also due to lacking mean (SD) for these markers [10,12,21,23,25]. Therefore, we evaluated the consistency of the results from the 9 studies with the study hypothesis (with regard to the direction of the association) at the marker level, subgroup level, and study level (Supplemental Table 4). Al-Daghri et al., 2015 reported no differences between participants with and without diabetes in urinary thiamine [21]. Adaikalakoteswari et al., 2012 [25] reported higher urine thiamine in people with diabetes and normoalbuminuria compared to the controls, while urinary thiamine was not higher in people with diabetes and albuminuria or microalbuminuria compared to the controls. In contrast, Thornalley et al., 2007 and Al-Attas et al., 2012 found higher excretion of thiamine in 24 h urine or spot urine in individuals with diabetes compared to those without diabetes [10,12].

Table 1

Brief description of the identified studies on the association between diabetes and thiamine markers.

First Author, year	DesignCountry	Study main question	Total n	% men; age, yrs	Inclusion criteria for patients	Main exclusion criteria for patients	Controls	Diabetes Type
Gonçalves et al., 2021 [23]	CC Brazil	Association between diabetes and thiamine deficiency in patients with covid-19	270	M 51.1 %; Median (IQR) = 74 (14) yrs	age > 18 yrs.; blood gases, free plasma thiamine and arterial lactate available within 72 h of admission; Covid19 +; respiratory failure	severe liver failure; CKD or dialysis; metformin use in the last 3 d	Covid(+) without diabetes	T2DM?
Cinici et al., 2020 [34]	CC Turkey	Association between thiamine and retinopathy in patients with T2DM	100	50 % M; mean (SD) = 62 (6.7) yrs	Patients with T2DM	alcoholism, uncontrolled hypertension, anemia, intestinal diseases, malnutrition, bariatric surgery, renal dysfunction, cancer, malabsorption disorders, vitamin use	Healthy; free of diabetes and retinopathy, age and sex matched	T2DM
Anwar et al., 2020 [33]	CC Pakistan	Association between diabetes and thiamine	90	Controls: 50 % M, 29 (8)yrs.; T1DM 27% M, 24 (6) yrs. T2DM 33%M, 43 (11) yrs	Diabetes	using diuretics, sig. Comorbidities, transplant surgery	healthy controls	T1DM+ T2DM
Nath et al., 2017 [35]	? CS USA	Prevalence of thiamine deficiency in obese patients admitted for bariatric surgery	400	15–17 % M; mean 45–47 yrs	age ≥ 18 yrs.; BMI > 34.9 kg/m ²	alcoholism, bariatric surgery, abnormal thyroid function	obese people admitted to the bariatric surgery without diabetes	T2DM?
Chalasova et al., 2017 [14]	CC Czech Republic	Association between TKA and diabetes and renal dysfunction	83	≈58 % M; median 53 to 68	T2DM ≥ 10 yrs.; GFR ≥ 60 l/min/1.73m ² or GFR 15–59 ml/min/1.73m ²	not clear?	healthy people; Ø renal disease verified by GFR	T2DM
Al-Daghri et al., 2015 [21]	? CC Saudi Arabia	Association of T1DM and thiamine markers	158	NR; mean 45 (18) yrs	participants in an ongoing study on biomarkers screening adults, T2DM for at least 5 yrs., HbA1c < 10 %, BMI 19–40 kg/m ²	not reported	healthy individuals	T1DM
Nix et al., 2015 [15]	CC Germany	vitamin B status, especially B6, in patients with type 2 diabetes with and without nephropathy	174	≈50 % M; med. 72 yrs. in patients and 43 yrs. in the controls	T2DM, with neuropathy, with normal renal function	ESRD, alcoholism, comorbidities, renal/pancreatic transplant, use of B vitamins, pregnancy and lactation	Healthy staff workers	T2DM
Michalak S et al., 2013 [27]	CC Poland	RBC-TKA in patients with diabetes or alcoholism and controls	49	72 % M in patients and 20%M in the control; mean 48–49 yrs	T2DM, with neuropathy, with normal renal function	nephropathy, CKD, renal insufficiency pancreatitis, liver dysfunction, vitamin supplementation	healthy individuals	T2DM
Al-Daghri et al., 2013 [11]	CS Saudi Arabia	Association of metabolic syndrome with thiamine status	236	70 % M; (SD) 49 (17) to 54 (13) yrs	Adults from a biomarker survey (no further details)	Earlier reported: ESRD, sig. Comorbidities, taking thiamine, transplant, pregnant and lactating women	normoglycemic people from the same survey	Hyperglycemia
Waheed et al., 2013 [13]	CC? Pakistan	Study levels of thiamine in patients with diabetes and the association with lipids	80	NR; Range 18–65 yrs	T2DM for ≥5 yrs., age 18–65 yrs., BMI 19–40 kg/m ² , HbA1c < 10 %; Albumin in urine <30 mg/24 h; 30–300 mg/24 h; and > 300 mg/24 h	chronic liver disease, ischemic heart disease, transplant surgery, ESRD, taking B vitamin suppl., pregnant and lactating women	age and sex matched healthy; fasting glucose <108 mg/dl	T2DM
Polizzi et al., 2012 [16]	CC Turkey	Study whether B1 is related to diabetes and the effect of B1 and B6 on glycation in the patients with	61	%NR; sex matched; 61 (10) yrs	T2DM, mean age 61 yrs. (45–75 yrs), not taking suppl.; albumin in urine	not described	age and sex matched healthy	T2DM

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Table 1 (continued)

First Author, year	DesignCountry	Study main question	Total n	% men; age, yrs	Inclusion criteria for patients	Main exclusion criteria for patients	Controls	Diabetes Type
Al-Attas et al., 2012 [10]	CC Saudi Arabia	diabetes and nephropathy Blood and urine thiamine status in patients with T1DM and T2DM with and without microalbuminuria	231	52 %; Range 18–67 yrs	<300 mg/24 h or > 300 mg/24 h Albumin excretion rate (ACR) normal <51 mg/g; micro-albuminuric when random urine ACR 51–391 mg/g; age (1–65 yrs), diabetes for ≥5 yrs., HbA1c <10 %, BMI 19–40 kg/m ²	ESRD (creatinine clearance <10 ml/min), allergy or intolerance to thiamine, thiamine suppl., pregnant or breastfeeding women	individuals without diabetes	T1DM and T2DM
Adaikalakoteswari et al., 2012 [25]	CC Indonesia	B-vitamin status in patients with T2DM with and without microalbuminuria and control individuals	152	≈50 % M; 52 (6) to 58 (4) yrs. (range 45–65 yrs)	T2DM, matched for age (45–65 yrs) and sex, diabetes for ≥5 yrs., HbA1c < 10 % and BMI 19–40 kg/m ² , 2 months washout from suppl. Albumin/creatinine ratio; < 22 and < 31 mg/g for M and F, respectively; 22–220 and 31–220 mg/g for M and F, respectively; and > 220 mg/g and plasma creatinine <2 mg/dl	plasma creatinine ≥2 mg/dl, liver diseases (verified markers), anemia, tuberculosis, severe CVD	healthy people, same social background as the patients	T2DM
Alam et al., 2012 [36]	RCT, CC Pakistan	Study the effect of high dose thiamine on thiamine levels, metabolic profile, microalbuminuria in patients with type 2 diabetes and to ascertain a therapeutic effect	60)	55%M; 52.7 (8.4) yrs.	T2DM, age 36–65 yrs., duration ≥5 yrs., BMI 19–40 kg/m ² , microalbuminuria defined as urine albumin 30 and 300 µg/min in 3 overnight urine collections	hepatitis, taking B1 suppl., alcoholism, pregnancy or lactation in women, severe hypertension	good health, matched for age and gender to the patients	T2DM
Satyanarayana et al., 2011 [17]	CC India	The status of B-vitamins and homocysteine in diabetic retinopathy	135	NR	T2DM with and without retinopathy	Control and diabetes individuals on nutritional supplements for the last 6 months and those with a history of nephropathy or other complications	partners, relatives, and friends of patients and employees, all asymptomatic, age ≥ 50 yrs. without CVD or renal disorders	T2DM
Agte and Tarwadi, 2008 [30]	CC India	Differences in micronutrient and antioxidant statuses between patients with cataract and diabetes vs. diabetes mellitus and cataract alone	319	52 % M; 60 (3)	Diabetes duration ≤2 yrs., age 50–70 yrs	NR	healthy people from the accompanying patients	T2DM
Thornalley et al., 2007 [12]	CC UK	Assess plasma, erythrocytes and urine thiamine in patients with type 1 and type 2 diabetes	94	52 %; 48 (15) yrs	Patients with diabetes; Albumin excretion rate: <30 mg/24 h or 30–300 mg/24 h; matched for age and sex (18–65 years); diabetes for ≥5 yrs.; HbA1c <10 %; and BMI 19–40 kg/m ²	ESRD; alcoholism; sig. Comorbidities; allergy or intolerance to thiamine; use of thiamine; organ transplants; pregnancy and lactation	healthy volunteers from partners and friends of the patients and investigators	T1DM and T2DM
Jermendy, 2006 [32]	CC Hungary	Assess RBC-TKA in diabetes patients	135	52 %; 4 (SEM 1,3) yrs	NR	serum creatinine >150 µM, heart failure, alcoholism, veganism, ketoacidosis	healthy individuals	T1 + T2DM
Agte et al., 2004 [29]	CC India	Assess RBC-TKA in patients with T2DM	65	54 %; 53.7 (14.7) yrs	T2DM, no further details	NR	age-, sex, socio-economically matched controls	T2DM
Wilkinson et al., 2000 [18]	CS New Zealand	To compare RBC-TTP levels in elderly	322		History	People in permanent nursing homes providing	100 young blood donors, 202	DM not specified

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Table 1 (continued)

First Author, year	DesignCountry	Study main question	Total n	% men; age, yrs	Inclusion criteria for patients	Main exclusion criteria for patients	Controls	Diabetes Type
Havivi et al., 1991 [31]	CC Israel	people with and without diabetes To assess RBC-TKA in patients with T2DM	212	49 %; 57 (11) yrs	NR	hospital levels of care NR	elderly without DM by history healthy individuals	T2DM
Kjos and Seim, 1977 [26]	CC Norway	To assess RBC-TKA in patients with T2DM and T1DM	44	NR	NR	NR	healthy hospital staff	T1 + T2DM
Watson and Dako, 1975 [28]	CC Ghana	To assess RBC-TKA in patients with T2DM and T1DM	177	67 %; age NR	NR	NR	medical exam at polyclinic, symptom free staff members	T1 + T2DM
Haugen, 1964 [24]	CC Norway	Study thiamine concentration in patients with diabetes	98	age T2DM 59–76 yrs.; T1DM 19–63 yrs	NR	NR		T2DM

BMI, body mass index; CC, case control study; CKD, chronic kidney disease; CVD, cardiovascular disease; HbA1c, glycated hemoglobin; ESRD, end stage renal disease; NR, not reported; RBC, red blood cells; RCT, randomized clinical trial; T1DM, type 1 diabetes mellitus; T2DM, type 2 diabetes mellitus; TKA, transketolase activity.

Table 2

Association between the presence of diabetes and thiamine concentrations.

Reference	SMD (95%CI)	Diabetes (n)	Controls (n)	Weight (%)
Thiamine				
Al-Attas et al., 2013 [10]	−0.84 (−1.15, −0.53)	205	52	14.5
Al-Daghri et al., 2013 [11]	−0.20 (−0.49, 0.09) ^a	171	65	14.5
Thornalley et al., 2007 [12]	−4.31 (−4.98, −3.63)	74	40	13.6
Waheed et al., 2013 [13]	−2.35 (−2.81, −1.88)	60	60	14.2
Chalasova et al., 2017 [14]	0.25 (−0.23, 0.73)	59	24	14.2
Nix et al., 2015 [15]	0.71 (0.37, 1.04)	122	52	14.5
Al-Daghri et al., 2015 [21]	−0.27 (−0.59, 0.04)	77	81	14.5
Pooled SMD and 95% CI ^b	−0.97 (−1.89, −0.06)			

CI, confidence intervals; n, numbers; SMD, standardized mean difference.

^a An SD value of 2.0 for thiamine was imputed for both study groups instead of the originally reported values of 0.04 and 0.06 for hyperglycemic and normoglycemic individuals, respectively.

^b Random effect model, $n = 7$ studies; $I^2 = 97.62$, $P < 0.001$; $\text{Tau}^2 = 1.47$ (SE = 0.95).

Regarding blood markers, Cinici et al., 2020 reported a higher percentage of individuals with diabetes (with and without retinopathy) with lower whole blood TPP compared to control individuals [34]. Anwar et al., 2020 reported markedly lower serum thiamine in individuals with type 1 and type 2 diabetes compared to the controls [33]. Nath et al., 2017 reported either low thiamine status in 17.8 % of the individuals with diabetes versus 15.7 % of the controls [35]. Alam et al., 2012 reported lower median plasma thiamine in individuals with diabetes compared to the controls [36]. Thus, the majority of these studies are in line with the results of the present meta-analysis.

The sensitivity analysis for thiamine, TMP, TDP, and total thiamine with the study of Al-Daghri et al., 2013 [11] entered in the data analysis using the originally reported standard deviations are shown in Supplemental Table 5.

3.2.7. Diabetes and the prevalence of thiamine deficiency

Three studies reported on the prevalence of thiamine deficiency among people with diabetes compared to control subjects. However, due to the heterogeneity in defining thiamine deficiency, we did not combine the results in a meta-analysis. Nath et al. reported that low thiamine (defined as either low whole blood thiamine or clinically

relevant deficiency that was improved after B1 treatment) was present in 17.8 % of people with diabetes (27 out of 152) and 15.7 % of the controls (39 out of 248) [35]. Gonçalves et al. measured whole blood TDP and defined low thiamine concentrations as $< 28 \mu\text{g/l}$ (or $< 78 \text{ nmol/l}$) [23]. These authors reported that 26.3 % of the patients versus 7.7 % of the controls had low whole blood TDP concentrations. Cinici et al. measured whole blood TDP (cut-off not reported, we estimated the cut-off as $< 65 \text{ nmol/l}$) and showed that 67.5 % of the patients with diabetes and 15.0 % of the controls had low whole blood thiamine [34].

3.2.8. Study quality and risk of bias evaluation

As shown in Supplemental Tables 6 to 8, many of the studies were at high risk of bias. The most common bias was related to selection of individuals with diabetes and controls. In addition, several studies have possible confounding due to between-group differences in age and sex of the participants. The description of pre-analytical conditions, analytical methods used for measurements of the biomarkers, and analytical accuracy was either insufficient or completely missing in the majority of publications. High risk of bias due to selection of the control individuals (i.e., younger than the individuals with diabetes) could have an impact on the generalizability of the results. Furthermore, our results suggest that albuminuria is a potential confounder that needs to be adjusted for in future studies on the association between diabetes and thiamine.

4. Discussion

The present systematic review and meta-analysis showed lower systemic concentrations of thiamine, TMP, and total thiamine in individuals with diabetes compared to those without diabetes, while TDP and erythrocyte transketolase activity tended to be lower in the diabetes group without reaching statistical significance. Subgroup analysis indicated that thiamine concentrations were lower in individuals with diabetes and albuminuria as compared to controls. Despite the heterogeneity between studies and possible selection bias in some of the studies, the results were consistent across the different thiamine markers, suggesting that requirement for thiamine may be higher in individuals with diabetes compared to those without diabetes.

4.1. Putative mechanisms of lower thiamine levels in diabetes

The possible reasons for the lower levels of thiamine markers in people with diabetes as observed herein are not well understood, but several mechanisms may be relevant. Thiamine deficiency leads to a pronounced impairment in islet cell insulin synthesis and secretion [37,38]. Conversely, insulin-deficient rats show reduced intestinal absorption of free thiamine and TMP [39]. These findings suggest a bidirectional effect, whereby insulin deficiency may elicit thiamine

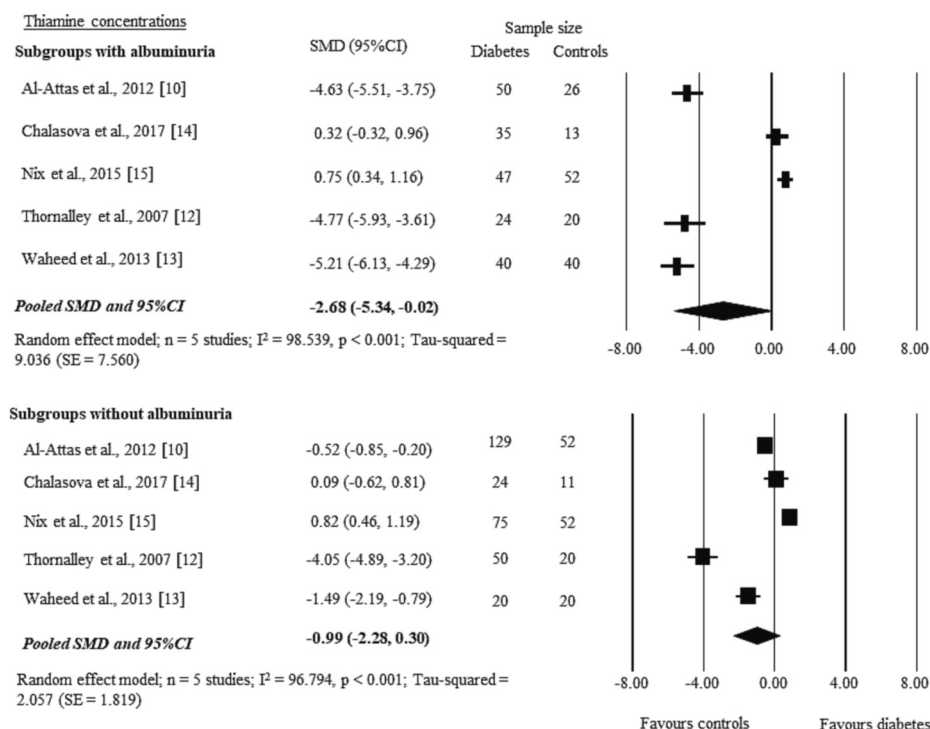


Fig. 2. Forest Plot for the association between concentrations of thiamine and diabetes according to the presence and the absence of albuminuria. Chalasova k et al., defined chronic kidney disease (CKD) 1–2 as ($GFR \geq 60$ ml/min/1.73 m²) and CKD 3–4 as ($GFR 59$ – 15 ml/min/1.73 m²); the controls had GFR values ≥ 90 ml/min/1.73 m² (using CKD-EPI formula).

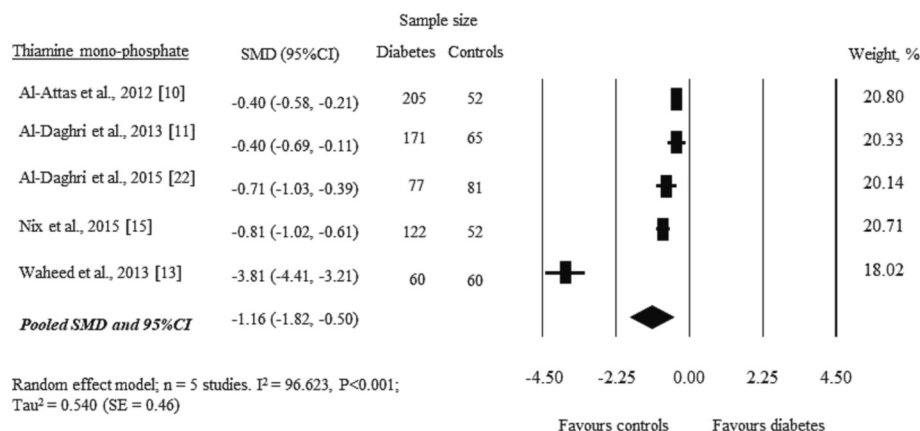


Fig. 3. Forest Plot for the association between the concentrations of thiamine monophosphate (TMP) in plasma, serum or whole blood (WB) and the presence of diabetes.

deficiency and vice versa. In line with this suggestion, higher dietary thiamine intake was associated with a lower odds ratio of the metabolic syndrome in a cohort with multiple cardiovascular risk factors [40]. In contrast, a systematic review and meta-analysis showed that thiamine treatment did not improve HbA1c or plasma glucose in persons with type 2 diabetes [41]. However, the number of available studies was low, the duration of treatment was short, while the daily thiamine dose and the treated patients showed high biometrical variability. Therefore, better designed studies are needed to investigate whether thiamine supplementation may improve diabetes control.

We found that the association between diabetes and lower thiamine concentrations was more pronounced in individuals with diabetes and albuminuria compared to those without albuminuria. In theory, altered renal thiamine handling in individuals with diabetes could contribute to thiamine deficiency leading to increased renal clearance of thiamine,

possibly due to a decreased reuptake of thiamine in renal proximal tubules [12]. Reuptake of thiamine takes place in the proximal tubules by thiamine transporters THTR-1 and THTR-2. These transporters are regulated by transcription factor specificity protein 1 (Sp1). In the hyperglycemic state, enhanced hexosamine pathway activity leads to enhanced O-glycosylation of the SP1 possibly altering thiamine transport in the proximal tubules [42]. Hexosamine-induced inhibition of THTR-1 and -2 by dicarbonyl glycation and tubular lumen acidification could be the mechanism responsible for the impairment of tubular reuptake of thiamine [12,43]. However, a previous study reported increasing plasma levels of thiamine, TMP, and TDP increase in parallel to the decline in renal function from micro- to macroalbuminuria and end stage renal disease [44]. Likewise, transketolase activity and TDP in erythrocytes gradually increased with decreasing kidney function [14]. Thus, it is conceivable that a biphasic course of thiamine status may

Table 3

Association between the presence of diabetes and concentrations of thiamine diphosphate (TDP) and total thiamine.

Reference	SMD (95%CI)	Diabetes (n)	Controls (n)	Weight (%)
Thiamine diphosphate				
Al-Attas et al., 2012 [10]	−5.43 (5.98, −4.87)	205	52	10.8
Al-Daghri et al., 2013 [11]	−0.99 (−1.29, −0.69) ^a	171	65	11.2
Chalasova et al., 2017 [14]	0.01 (−0.46, 0.49)	59	24	11.0
Nix et al., 2015 [15]	0.69 (0.36, 1.02)	122	52	11.3
Polizzi et al., 2012 [16]	0.48 (−0.03, 0.99)	31	30	10.9
Saathanarayanan et al., 2011 [17]	0.31 (0.01, 0.60)	90	90	11.2
Wilkinson et al., 2000 ^b [18]	−0.37 (−0.71, −0.03)	38	302	11.2
Al-Daghri et al., 2015 [21]	−0.55 (−0.87, −0.23)	77	81	11.2
Gonçalves et al., 2021 [23]	−0.72 (−0.97, −0.47)	114	156	11.3
Pooled SMD and 95% CI	−0.72 (−1.54, 0.11) ^c			
Total thiamine				
Al-Attas et al., 2012 [10]	−1.71 (−2.05, −1.37)	205	52	24.9
Al-Daghri et al., 2013 [11]	−0.90 (−1.20, −0.61) ^a	171	65	25.7
Al-Daghri et al., 2015 [21]	−0.59 (−0.91, −0.27)	77	81	25.3
Haugen, 1964 [24]	−0.85 (−1.23, −0.47)	37	122	24.1
Pooled SMD and 95%CI	−1.01 (−1.48, −0.54) ^d			

CI, confidence intervals; n, numbers; SMD, standardized mean difference.

^a We conducted the analysis after imputing a standard deviation value of 15.0 for TDP and 15.5 for total thiamine for the study of Al-Daghri et al., 2013 [11].

^b The study of Wilkinson et al. [18] was included with 302 controls (we combined the elderly and the young subgroups of the controls).

^c Random effect model; $n = 9$ studies; $I^2 = 98.04$, $p < 0.001$; $\text{Tau}^2 = 1.54$ (SE = 0.74).

^d Random effect model; $n = 4$ studies. $I^2 = 87.63$, $p < 0.001$; $\text{Tau}^2 = 0.20$ (SE = 0.19).

occur in individuals with diabetes starting with normal or decreased concentrations followed by increased levels in advanced kidney disease, but this hypothesis should be verified in appropriate longitudinal studies. Apart from nephropathy, the thiamine levels in other diabetic microvascular complications such as neuropathy or retinopathy could not be assessed herein due to a lack of appropriate data.

Thiamine is a cofactor for transketolase that links the non-oxidative part of the pentose phosphate pathway and the oxidative decarboxyl metabolism of glucose. An open question is whether low thiamine in individuals with diabetes could be involved in the development and progression of diabetic microvascular complications. The pathophysiology of hyperglycemia-induced diabetic microvascular complications has been characterized by an increased flux in four major biochemical pathways including the polyol and hexosamine pathways, protein kinase C activation, and advanced glycation end-product formation induced by overproduction of superoxide by the mitochondrial electron-transport

chain. [4]. Superoxide overproduction partially inhibits the glycolytic enzyme glyceraldehyde phosphate dehydrogenase and thereby diverts upstream metabolites from glycolysis into glucose-driven signaling pathways of glucose overuse. Two of these upstream metabolites include fructose-6-phosphate and glyceraldehyde-3-phosphate which are also end products of the non-oxidative branch of the pentose phosphate pathway (PPP) and are generated by the enzyme transketolase that catalyzes several key reactions of the nonoxidative branch of the PPP [5,7]. Thiamine and its lipid-soluble derivative benfotiamine have been shown to normalize the aforementioned four metabolic pathways, reduce oxidative stress, and thereby prevent hyperglycemia-induced microvascular damage in animal models [5,7,8].

4.2. Clinical trials in diabetic microvascular complications

Several randomized clinical trials (RCTs) were conducted addressing the question as to whether treatment with thiamine or its derivatives would have favorable effects on diabetic polyneuropathy and nephropathy. Unlike thiamine, benfotiamine is absorbed passively and readily traverses the intestinal barrier, resulting in considerably higher plasma, blood, and erythrocyte concentrations than thiamine [45]. Treatment with benfotiamine has been shown to improve neuropathic symptoms within 3–6 weeks in individuals with diabetic polyneuropathy [46,47]. A randomized, double-blind, placebo-controlled parallel-group 1-year trial (BOND Study) to assess the effects of treatment with benfotiamine on morphometric, neurophysiological, and clinical measures in participants with type 2 diabetes and mild to moderate symptomatic polyneuropathy is currently underway [48]. In individuals with incipient diabetic nephropathy treated over 12 weeks, one RCT showed that urinary albumin excretion was reduced following thiamine treatment [49]. In contrast, another RCT using benfotiamine did not demonstrate such an effect [50]. Longer term RCTs at earlier time points, especially in individuals with low thiamine levels, are needed to determine whether diabetic microvascular complications may be ameliorated by thiamine or benfotiamine treatment.

4.3. Study limitations

The present study has some limitations. First, the search was restricted to articles published in English or German language in PubMed and the Cochrane Central Register of Controlled Trials. Although publication bias was not reported for most of the associations, the quality of the suitable studies was relatively poor. In particular, one study [11] reported implausibly low SDs for four thiamine markers. To minimize bias, we imputed the SDs from the corresponding SDs reported in an independent study by the same authors [21]. Second, the clinical utility of several thiamine markers is questionable. For example, the transketolase activity and activity coefficient are subject to methodological limitations causing high analytical variations between laboratories and between samples. Third, genetic variability in the gene encoding transketolase may represent another potential source of bias [51]. Fourth, differences in laboratory methods and pre-analytic conditions may increase the between-study variation (heterogeneity). However, since samples of patients and controls from each of the studies were measured by identical methods and since the effect size is the SMD, the overall association between diabetes and thiamine concentrations is not likely to be subject to bias due to analytical methods. Methods of measurement of transketolase activity are not standardized and could be subject to higher variations between laboratories, which may have attenuated the overall association. Finally, concentrations of thiamine compounds are lower in plasma compared to whole blood and are influenced by recent thiamine intake. For routine investigation of thiamine status, whole blood TDP appears to be the most suitable marker, because it is the coenzyme form of thiamine within the cells and its concentrations are high in whole blood, thus enabling a better analytical performance. The direction of the relationship between diabetes and

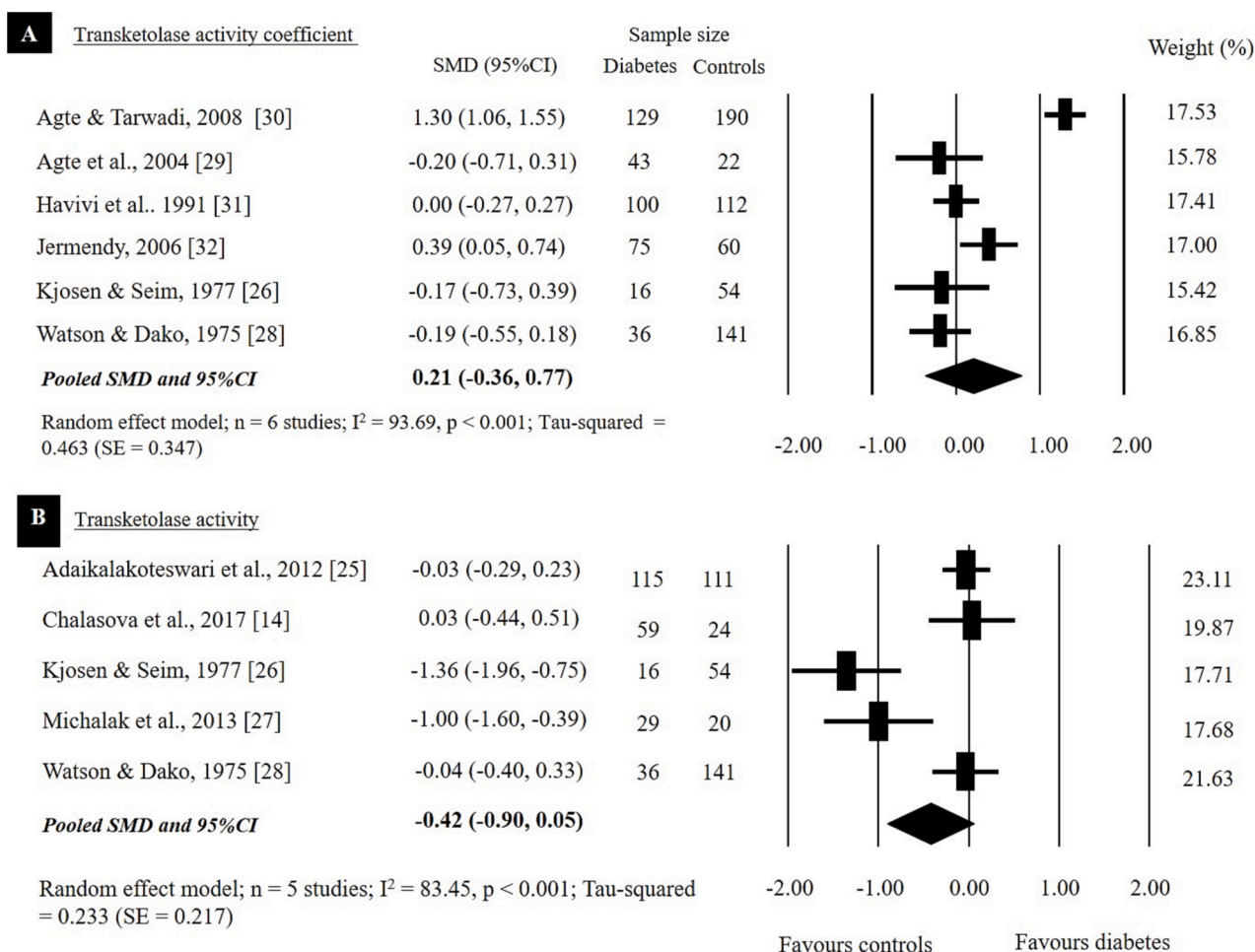


Fig. 4. Forest Plot for the association between A) transketolase activity coefficient (ETKAC) or B) transketolase activity and the presence of diabetes mellitus.

thiamine status was identical (inverse) for all five markers studied herein, but statistical significance was not reached for TDP and transketolase activity which could be due to the heterogeneity of the study populations or analytical aspects. On the other hand, it is also conceivable that TDP concentrations are maintained at the expense of thiamine and TMP. However, it is not known whether there is a specific biochemical role for TMP other than being an intermediate thiamine compound between free thiamine and TDP and TTP forms [52]. Thus, well-designed prospective studies should clarify whether the concentrations of thiamine markers differ in their temporal response to thiamine depletion.

5. Conclusions

Our results point to an association between diabetes and lower systemic thiamine markers including thiamine, TMP, and total thiamine compounds. Furthermore, individuals with diabetes and albuminuria could be particularly prone to lower thiamine concentrations. However, due to the overall low quality of the studies and possible bias, the present findings require confirmation by better designed future studies considering the numerous aforementioned concerns and possible confounders. While blood TDP appears to be the most important and reliable thiamine analyte, our results do not suggest that diabetes is unequivocally associated with low TDP levels. Thus, more research is needed to determine the suitability of TDP as a single marker compared to the other thiamine markers for routine clinical measurement and monitoring. In addition, clarifying the role of treatment and prevention with thiamine and its derivatives in diabetes-associated microvascular complications merits

further clinical trials.

Credit authorship contribution statement

DZ and RO conducted the search, data extraction and assessed the risk of bias and study quality. RO conducted data analyses. DZ and RO wrote the manuscript. All authors contributed to the protocol and discussion and reviewed and edited the manuscript. DZ and RO are the guarantors of this work and, as such, had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

Funding and the role of the sponsor

This study was financed by Wörwag Pharma, Böblingen, Germany. The sponsor had no role in concept and design of the study, search and screening of the literature, data extraction, data analysis, interpretation, or drafting of the manuscript.

Declaration of competing interest

DZ, KR, AS, and RO received compensation for scientific consulting and speaking activities from Wörwag Pharma. DZ and AS received research grants from Wörwag Pharma.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.metabol.2023.155565>.

[org/10.1016/j.metabol.2023.155565](https://doi.org/10.1016/j.metabol.2023.155565).

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